Vinicius Arruda

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Experience

- **QRA Corp.** — Halifax, Canada (Remote)
  Machine Learning Engineer — *Oct 2022 – Present*
  - Designed and built NLP/LLM-based software to analyze, generate, and manage complex engineering requirements across large document collections.
  - Implemented retrieval-augmented generation (RAG) pipelines to ground model outputs in proprietary engineering documentation and source code.
  - Developed LLM-based agents to rewrite, validate, and generate software requirements from technical documents and codebases.
  - Combined classical NLP techniques (POS tagging, dependency parsing, rule-based analysis) with LLMs to improve reliability and controllability of outputs.
  - Worked with both open-source and proprietary models, including GPT-5, Mixtral, LLaMA, Falcon, and Mistral.
  - Led technical deep-dive presentations with an enterprise stakeholder, contributing to securing a $2M contract.
  - Took ownership across the full lifecycle, from experimentation and evaluation to proof-of-concepts and production.
  - Tech: Python, Large Language Models, NLP, RAG, OpenAI API, LangChain, LangGraph, Agents SDK, SpaCy, Ollama, FAISS.
- **Samsung Institute for Development in Informatics** — Campinas, Brazil (Remote)
  Machine Learning Engineer — *Jul 2021 – Oct 2022*
  - Contributed to the end-to-end text-to-speech (TTS) pipeline for Samsung products, covering text normalization, grapheme-to-phoneme conversion, POS tagging, acoustic modeling, and vocoding.
  - Trained, evaluated, and improved models including Tacotron, LPCNet, HMMs, and CART for Brazilian Portuguese (pt-BR) language.
  - Designed and maintained text normalization rules used in Samsung Bixby's text-to-speech pipeline.
  - Built and maintained linguistic resources, including corpora and lexicons.
  - Developed a text anonymization pipeline using named entity recognition (NER) to support data privacy requirements.
  - Collaborated closely with global teams across Poland, South Korea, and Greece.
  - Identified and fixed a critical issue in Samsung's Bixby TTS pipeline, resolving a bug that affected users globally.
  - Tech: Python, PyTorch, TensorFlow, C++, SpaCy, NLTK, Docker.
- **Visialabs** — Vitória, Brazil
  Machine Learning Engineer — *Nov 2020 – Jul 2021*
  - Trained, evaluated, and deployed real-time object detection models (EfficientDet, YOLOv3) into production.
  - Built multi-camera inference pipelines, managing concurrent video streams with inter-process communication.
  - Optimized models for low-latency, real-time inference, for deployment on resource-constrained hardware such as NVIDIA Jetson Nano.
  - Integrated ML inference with backend and frontend systems, supporting end-to-end product workflows.
  - Collected, curated, and annotated image datasets to train and evaluate object detection models.
  - Tech: Python, PyTorch, TensorFlow, OpenCV, Pillow, Docker.
- **CNPq** — Vitória, Brazil
  Research Fellow — *Mar 2019 – Nov 2020*
  - Conducted funded research in deep learning and computer vision, resulting in peer-reviewed publications.
  - Researched unsupervised domain adaptation, synthetic data generation, style transfer, and object detection.
  - Worked across theory and implementation, translating research ideas into practical experiments.
  - Tech: Python, PyTorch, TensorFlow, OpenCV, GANs, CycleGAN, Faster R-CNN, EfficientDet, YOLO.
- **Olho do Dono** — Vitória, Brazil
  Machine Learning Engineer Intern — *Mar 2018 – May 2018*
  - Implemented and evaluated semantic segmentation models for cattle weight estimation using stereo images.
  - Contributed to early-stage ML product development in an applied, real-world setting.

- Tech: Python, PyTorch, TensorFlow, OpenCV, Detectron.

- **FAPES** Vitória, Brazil
  Research Assistant *May 2015 - Nov 2016*
  - Conducted funded research in machine learning and metaheuristics, applied to time series prediction for stock market forecasting, resulting in a peer-reviewed publication.
  - Tech: Python, MATLAB, C/C++, Scikit-learn.

Education

- **Universidade Federal do Espírito Santo** Vitória, Brazil
  Master of Science in Computer Science; GPA 9.38/10.0 *Mar 2019 – Feb 2022*
  - Dissertation: Cross-Domain Object Detection Using Unsupervised Image Translation and Neural Style Transfer.
- **Universidade Federal do Espírito Santo** Vitória, Brazil
  Bachelor in Computer Science; GPA 9.06/10.0 *Mar 2014 – Dec 2018*
  - Thesis: Improving Cross-Domain Object Detection Using Unsupervised Image-to-Image Translation: Car Detection in Day to Night Images.

Publications

- Cross-Domain Object Detection Using Unsupervised Image Translation
  Expert Systems with Applications (ESWA) *2022*
- Cross-Domain Car Detection Using Unsupervised Image-to-Image Translation: From Day to Night
  International Joint Conference on Neural Networks (IJCNN) *2019*
- Particle Swarm Optimization for training artificial neural networks of type ELM: A case study for time series prediction
  XLVIII Brazilian Symposium of Operational Research (SBPO) *2016*

Scholarships

- **2019**: Full scholarship from Brazil's research agency CNPq (R$ 18,000/year) for a two-years master's degree program at the Universidade Federal do Espírito Santo.
- **2015**: The State of Espírito Santo Research Agency (FAPES) scholarship for undergraduate research (R$ 4,800/year).

Services

- **Social chair:**
  - **2021**: LatinX in AI Social at AAAI 2021. *(www.latinxinai.org/aaai-2021)*
- **Teaching assistant:**
  - **2020**: Software Development in C, UFES. *(Instructor: Vinicius Passos)*
  - **2020**: Software Development in Python, UFES. *(Instructor: Vinicius Passos)*